GENETICS · MCQ SET

Genetic Disorders of the Urinary Tract

LG 1
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GUE Block

Genetics GUE Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, 5 options each (A–E), mixed difficulty, 16 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy ● 10 Medium ● 6 Hard

1. EASY

The collecting system of the kidney develops from which embryonic structure?

A. Metanephric mesenchyme
B. Ureteric bud
C. Paramesonephric duct
D. Neural crest
E. Allantois

2. EASY

Which gene is associated with autosomal recessive polycystic kidney disease (ARPKD)?

A. PKD1
B. COL4A5
C. NPHP1
D. PKHD1
E. GREB1L

3. EASY

According to the lecture, nephronophthisis is inherited in which pattern?

A. Autosomal recessive
B. Autosomal dominant
C. X-linked
D. Mitochondrial
E. Y-linked

4. EASY

Alport syndrome is a disorder of which structure?

A. Polycystin signalling in tubular cells
B. Type I collagen in bone and skin
C. Type IV collagen of basement membranes
D. Ciliary proteins of the renal tubule
E. Fibrillin of elastic fibres

5. MEDIUM CASE

A term newborn needs ventilation from birth. Prenatal ultrasound showed absent kidneys and renal arteries, a non-visualised bladder and almost no amniotic fluid.

Which sequence explains her respiratory failure?

A. Pyelonephritis → renal scarring → hypertension → pulmonary oedema
B. Hepatic fibrosis → portal hypertension → ascites → splinting
C. Reflux → febrile UTI → sepsis → respiratory failure
D. Cyst growth → huge kidneys → diaphragm splinting → hypoxia
E. Anuria → oligohydramnios → pulmonary hypoplasia

6. MEDIUM CASE

A 30-year-old woman has an ultrasound for unrelated pain. It shows no left kidney and an enlarged right kidney. She is well.

Besides blood pressure and albuminuria over time, what does the lecture say to look for?

A. Congenital hepatic fibrosis
B. Genital tract anomalies
C. Lenticonus
D. Polycystic liver disease
E. Sensorineural hearing loss

7. MEDIUM CASE

A 3-year-old girl has had two febrile UTIs with pyelonephritis. A voiding study shows urine flowing back up both ureters.

Which anatomical feature begins the chain that the lecture describes?

A. Posterior urethral valves
B. Pelvi-ureteric junction (UPJ) obstruction
C. Horseshoe kidney (a position anomaly)
D. Short intramural ureteric tunnel
E. Duplex collecting system

8. MEDIUM CASE

The parents of the girl with VUR ask whether her baby brother is at risk.

Which statement about the genetics of VUR does the lecture support?

A. Familial aggregation is common; most isolated VUR is genetically complex
B. VUR is X-linked, so only the boys in this family can ever be affected
C. VUR is autosomal recessive, with a 1 in 4 risk for each of her siblings
D. VUR is caused by PKD1 variants in most of the families it affects
E. VUR has no familial tendency at all, so her siblings need no thought

9. MEDIUM CASE

A 4-month-old boy has bilateral renal hypoplasia, a congenital heart defect and hearing loss.

On the lecture's testing ladder, which test addresses step 2 for him?

A. PKHD1 sequencing on its own
B. Testing both parents instead of the child
C. Chromosomal microarray or validated CNV calling
D. No testing until a second child is affected
E. Targeted COL4A5 sequencing

10. MEDIUM CASE

A 28-year-old man has BP 152/94, normal eGFR and bilateral renal cysts. His father reached kidney failure at 54.

Which counselling statement fits the lecture?

A. Each child has a 1 in 4 risk, because both parents must be carriers
B. Each child has a 50% risk; relatives may choose not to know their status
C. The risk depends on the child's sex, because the disease is X-linked
D. His children are not at risk while his eGFR stays normal
E. All his siblings must be tested, whether or not they wish to know

11. MEDIUM CASE

A 45-year-old woman with ADPKD and many liver cysts is reviewed in clinic. Her uncle died of a brain haemorrhage.

Which additional assessment does the lecture say selected ADPKD patients need?

A. Slit-lamp examination for lenticonus
B. Audiometry for hearing loss
C. Liver biopsy for hepatic fibrosis
D. Chromosomal microarray
E. Intracranial aneurysm risk assessment

12. MEDIUM CASE

Prenatal ultrasound showed enlarged echogenic kidneys and reduced amniotic fluid. The newborn has respiratory compromise. The parents are healthy first cousins.

What is the most likely diagnosis, and which gene?

A. ARPKD — PKHD1
B. ADPKD — PKD1
C. Alport syndrome — COL4A5
D. Nephronophthisis — NPHP1
E. Renal agenesis — GREB1L

13. MEDIUM CASE

A 14-year-old boy has had microscopic hematuria since age 6, now with albuminuria. Complement is normal. He has high-frequency hearing loss; his mother has hematuria.

Which genetic test does the lecture recommend?

A. PKD1 and PKD2 sequencing plus deletion/duplication analysis
B. Chromosomal microarray alone, without sequencing
C. PKHD1 sequencing followed by a liver biopsy
D. COL4A3/4/5 sequencing plus deletion/duplication analysis
E. A karyotype of the boy and both his parents

14. MEDIUM CASE

Exome sequencing for a 2-year-old with renal dysplasia reports a variant of uncertain significance (VUS) in HNF1B.

How should this result be handled?

A. Treat it as the confirmed diagnosis and stop any further work-up
B. Report it to the family as a negative result, since it is not pathogenic
C. Weigh ACMG class, phenotype fit and segregation — not a diagnosis
D. Offer prenatal testing for it in the next pregnancy
E. Ignore it, because HNF1B has no role in the kidney

15. HARD CASE

A woman carries her family's COL4A5 variant and has persistent microscopic hematuria. Her father reached kidney failure at 42 and her son has Alport syndrome. A relative calls her "just a benign carrier".

Which response fits the lecture?

A. That is correct — carrier females in X-linked disease never get kidney disease
B. That is correct — her hematuria must have another cause
C. Her disease is autosomal dominant, so the term does not apply
D. Her sons' risk is 1 in 4, whatever their sex
E. Avoid the term — severity differs by sex and variant, and she already has hematuria

16. HARD CASE

An embryo carries a variant that abolishes GDNF → RET signalling at 4–5 weeks.

Which outcome is most expected?

A. No ureteric bud grows out, so nothing is induced — renal agenesis
B. Nephrons form normally, but the collecting ducts dilate — as in ARPKD
C. The glomerular basement membrane forms abnormally — as in Alport syndrome
D. Kidneys form normally, then fill with cysts in adult life — as in ADPKD
E. Nephrogenesis carries on after birth to make up the shortfall

17. HARD CASE

A 19-year-old has bilateral variable-sized kidney cysts and a pathogenic PKD1 variant. Both parents have normal kidney imaging, and neither carries the variant in blood.

What is the best estimate of the risk for his parents' next child?

A. 1 in 2 — the textbook autosomal dominant figure still applies unchanged
B. Low but not zero: likely de novo, yet parental mosaicism is possible
C. 1 in 4 — both parents must be silent carriers of the variant
D. Zero — a de novo variant can never recur in a later sibling
E. Sex-dependent — PKD1 lies on the X chromosome in this family

18. HARD CASE

A 9-year-old has enlarged echogenic kidneys with mildly reduced eGFR, plus splenomegaly and oesophageal varices. Both parents have normal kidney ultrasound.

Which feature most firmly points to ARPKD rather than ADPKD?

A. Involvement of both kidneys at once
B. Raised blood pressure on repeated readings
C. The presence of cysts within the kidneys
D. Portal hypertension from hepatic fibrosis
E. Progressive chronic kidney disease (CKD)

19. HARD CASE

A term newborn whose mother had oligohydramnios has an absent left kidney and a small echogenic right kidney, with no other obvious anomaly. The parents ask to be tested first, instead of the baby.

What is the best next genetic step?

A. Test only the parents, since their result will explain the baby's
B. Sequence PKD1 and PKD2 in the baby
C. Test the baby with chromosomal microarray plus a CAKUT panel or exome
D. Wait until the child is older, since CAKUT is not genetic
E. Karyotype both parents and stop there

20. HARD CASE

A 5-year-old with syndromic CAKUT had a sequencing-only exome three years ago, reported as negative.

Which interpretation is best?

A. The cause is not genetic, so testing the family is unnecessary now
B. The negative exome rules out every known CAKUT gene for good
C. Only the parents need the same sequencing-only exome repeated
D. The child should now be labelled as carrying a VUS instead
E. A copy-number variant may be missed; reanalysis could find a cause

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	**B**	**Ureteric bud** — Collecting system ← ureteric bud; nephrons ← metanephric mesenchyme. The two tissues induce each other.
2	**D**	**PKHD1** — ARPKD is caused by PKHD1. PKD1/PKD2 cause ADPKD, COL4A5 Alport syndrome, NPHP1 nephronophthisis.
3	**A**	**Autosomal recessive** — The lecture lists ARPKD and nephronophthisis as the autosomal recessive examples, with a 1 in 4 risk per pregnancy.
4	**C**	**Type IV collagen of basement membranes** — Alport syndrome is a collagen IV basement-membrane disorder — abnormal α3α4α5 type IV collagen.
5	**E**	**Anuria → oligohydramnios → pulmonary hypoplasia** — Bilateral renal agenesis: no kidneys means no urine, too little amniotic fluid and underdeveloped lungs.
6	**B**	**Genital tract anomalies** — Unilateral agenesis: look for contralateral CAKUT, genital tract anomalies and hypertension/albuminuria over time. The enlarged kidney is compensatory hypertrophy.
7	**D**	**Short intramural ureteric tunnel** — VUR chain: short intramural ureteric tunnel → retrograde urine flow → febrile UTI/ pyelonephritis → scarring risk.
8	**A**	**Familial aggregation is common; most isolated VUR is genetically complex** — The lecture: familial aggregation is common, but most isolated VUR is genetically complex; multiple genes may contribute.
9	**C**	**Chromosomal microarray or validated CNV calling** — Step 2 (copy number): chromosomal microarray or validated CNV calling when CAKUT is syndromic — as it is here, with heart and hearing involvement.
10	**B**	**Each child has a 50% risk; relatives may choose not to know their status** — This is the ADPKD case: 50% transmission risk, and the right not to know must be respected.
11	**E**	**Intracranial aneurysm risk assessment** — Beyond the kidney, ADPKD causes liver cysts, and selected patients need intracranial aneurysm risk assessment.
12	**A**	**ARPKD — PKHD1** — Enlarged echogenic kidneys before birth with neonatal respiratory compromise fit ARPKD (PKHD1). Consanguinity raises the prior probability of a recessive disorder.
13	**D**	**COL4A3/4/5 sequencing plus deletion/duplication analysis** — Suspected X-linked Alport syndrome: COL4A3/4/5 sequence + deletion/duplication analysis.
14	**C**	**Weigh ACMG class, phenotype fit and segregation — not a diagnosis** — Step 4 of the ladder: ACMG classification + phenotype fit + segregation — a VUS is not a diagnosis.
15	**E**	**Avoid the term — severity differs by sex and variant, and she already has hematuria** — X-linked: transmission and severity differ by sex and variant; the lecture says to avoid the term "benign carrier".
16	**A**	**No ureteric bud grows out, so nothing is induced — renal agenesis** — GDNF → RET promotes ureteric bud formation and outgrowth. Without the bud, reciprocal induction fails, so no collecting system or nephrons form.
17	**B**	**Low but not zero: likely de novo, yet parental mosaicism is possible** — The lecture: always modify the textbook risk for de novo variation, parental mosaicism and the actual molecular result.
18	**D**	**Portal hypertension from hepatic fibrosis** — The two separate by pattern, not cysts alone: ADPKD gives liver cysts, ARPKD congenital hepatic fibrosis with portal hypertension. The other features occur in both.
19	**C**	**Test the baby with chromosomal microarray plus a CAKUT panel or exome** — Case 1: test the infant — not only the parents — and pair sequence analysis with copy-number detection. Parental imaging and segregation then clarify recurrence.
20	**E**	**A copy-number variant may be missed; reanalysis could find a cause** — Negative ≠ non-genetic: syndromic CAKUT needs copy-number testing, the method must detect the relevant variant classes, and reanalysis matters as knowledge evolves.